1. A real estate company wants to develop a system that predicts house prices based on square footage, number of bedrooms, and location.
   Q: Identify the problem type and outline the step-by-step logic to solve it.

   A: Supervised Learning – Regression (because the output is a continuous value: house price).
   Step by Step Logic:
   1. Use data given, Clean the data by handling missing values and outliers.
   2. Encode categorical using OneHotEncoder or LabelEncoder.
   3. Scale numerical features using StandardScaler.
   4. Split the dataset into training and testing sets (e.g., 80% train, 20% test).
   5. Choose a regression model and train the model using the training data.
   6. Evaluate model performance using metrics like R^2 score on the test data.
   7. Fine-tune the model with GridSearch CV and then deploy the best model

2. A bank wants to build a model to detect fraudulent transactions by analyzing customer spending behavior and transaction history.
   Q: Identify the problem type and outline the step-by-step logic to solve it.

   A: Supervised Learning – Classification (because the output is a category: fraudulent or not fraudulent).
   Step by Step Logic:
   1. Use data given, Clean the data by handling missing values and outliers.
   2. Encode categorical using OneHotEncoder or LabelEncoder.
   3. Scale numerical features using StandardScaler.
   4. Split the data into training and testing sets.
   5. Choose a classification model and train it.
   6. Evaluate performance using metrics like ROC-AUC score and F1 Weighted Score
   7. Fine-tune using GridSearchCV and deploy the best model

3. A supermarket wants to segment its customers based on their shopping patterns to provide personalized promotions.
   Q: Identify the problem type and outline the step-by-step logic to solve it.

   A: Unsupervised Learning – Clustering (because there are no labels, and the goal is to group similar customers).
   Step by Step Logic:
   1. Use data given, Clean the data by handling missing values and outliers.
   2. Select relevant features and scale the features using StandardScaler.
   3. Choose a clustering algorithm.
   4. Determine the optimal number of clusters using methods like the Elbow Method or Silhouette Score.
   5. Train the model and assign cluster labels to customers
   6. Analyze each cluster and use insights to design personalized promotions.

4. A company wants to estimate an employee's salary based on their years of experience, job title, and education level.

   Q: Identify the problem type and outline the step-by-step logic to solve it.

   A: Supervised Learning – Regression (because the output is a continuous value: salary).

   Step by Step Logic:

   1. Use data given, Clean the data by handling missing values and outliers.
   2. Encode categorical using OneHotEncoder or LabelEncoder.
   3. Scale numerical features using StandardScaler.
   4. Split the dataset into training and testing sets (e.g., 80% train, 20% test).
   5. Choose a regression model and train the model using the training data.
   6. Evaluate model performance using metrics like R^2 score on the test data.
   7. Fine-tune the model with GridSearch CV and then deploy the best model

5. An email provider wants to automatically classify incoming emails as spam or not spam based on their content and sender details.

   Q: Identify the problem type and outline the step-by-step logic to solve it.

   A: Supervised Learning – Classification (because the output is a category: spam or not spam).

   Step by Step Logic:

   1. Use data given, Clean the data by handling missing values and outliers.
   2. Encode categorical using OneHotEncoder or LabelEncoder.
   3. Scale numerical features using StandardScaler.
   4. Split the data into training and testing sets.
   5. Choose a classification model and train it.
   6. Evaluate performance using metrics like ROC-AUC score and F1 Weighted Score
   7. Fine-tune using GridSearchCV and deploy the best model

6. A business wants to analyze customer reviews of its products and determine whether the sentiment is positive or negative.

   Q: Identify the problem type and outline the step-by-step logic to solve it.

   A: Supervised Learning – Classification (because the output is a category: positive or negative sentiment).

   Step by Step Logic:

   8. Use data given, Clean the data by handling missing values and outliers.
   9. Encode categorical using OneHotEncoder or LabelEncoder.
   10. Scale numerical features using StandardScaler.
   11. Split the data into training and testing sets.
   12. Choose a classification model and train it.
   13. Evaluate performance using metrics like ROC-AUC score and F1 Weighted Score
   14. Fine-tune using GridSearchCV and deploy the best model

7. An insurance company wants to predict whether a customer is likely to file a claim in the next year based on their driving history and demographics.
   Q: Identify the problem type and outline the step-by-step logic to solve it.
   A: Supervised Learning – Classification (because the output is a category: will file a claim or not).

Step by Step Logic:

1. Use data given, Clean the data by handling missing values and outliers.
2. Encode categorical using OneHotEncoder or LabelEncoder.
3. Scale numerical features using StandardScaler.
4. Split the data into training and testing sets.
5. Choose a classification model and train it.
6. Evaluate performance using metrics like ROC-AUC score and F1 Weighted Score
7. Fine-tune using GridSearchCV and deploy the best model

8. A streaming platform wants to recommend movies to users by grouping them based on their viewing preferences and watch history.
   Q: Identify the problem type and outline the step-by-step logic to solve it.
   A: Unsupervised Learning – Clustering (because there are no labels, and the goal is to group users based on similar behavior).

Step by Step Logic:

1. Use data given, Clean the data by handling missing values and outliers.
2. Select relevant features and scale the features using StandardScaler.
3. Choose a clustering algorithm.
4. Determine the optimal number of clusters using methods like the Elbow Method or Silhouette Score.
5. Train the model and assign cluster labels to customers
6. Analyze each cluster and use insights to design personalized promotions.

9. A hospital wants to predict the recovery time of patients after surgery based on their age, medical history, and lifestyle habits.
   Q: Identify the problem type and outline the step-by-step logic to solve it.**A:** Supervised Learning – Regression (because the output is a continuous value: recovery time of patients).

Step by Step Logic:

1. Use data given, Clean the data by handling missing values and outliers.
2. Encode categorical using OneHotEncoder or LabelEncoder.
3. Scale numerical features using StandardScaler.
4. Split the dataset into training and testing sets (e.g., 80% train, 20% test).
5. Choose a regression model and train the model using the training data.
6. Evaluate model performance using metrics like R^2 score on the test data.
7. Fine-tune the model with GridSearch CV and then deploy the best model

10. A university wants to predict a student's final exam score based on study hours, attendance, and past academic performance.

    Q: Identify the problem type and outline the step-by-step logic to solve it.

    A: Supervised Learning – Regression (because the output is a continuous value: final exam score).

Step by Step Logic:

8. Use data given, Clean the data by handling missing values and outliers.
9. Encode categorical using OneHotEncoder or LabelEncoder.
10. Scale numerical features using StandardScaler.
11. Split the dataset into training and testing sets (e.g., 80% train, 20% test).
12. Choose a regression model and train the model using the training data.
13. Evaluate model performance using metrics like R^2 score on the test data.
14. Fine-tune the model with GridSearch CV and then deploy the best model